

# Talha Zain

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## SUMMARY

AI and Data professional with 3+ years of hands-on experience building data-driven and machine learning systems. Strong background in data ingestion, data preprocessing, ETL/ELT pipelines, SQL, and Python, with experience designing reliable datasets for analytics and machine learning. Worked extensively on large language models (LLMs) such as GPT-3 and Llama, including data preparation, fine-tuning, and deployment for NLP and generative AI use cases. Experienced in data cleaning, transformation, validation, and analytical data modeling using medallion architecture. Comfortable working with cloud platforms, APIs, and batch processing pipelines, and collaborating with analytics and business teams to deliver scalable, production-ready AI and data solutions.

## EXPERIENCE

**BoolMind — Lahore**  
ML Engineer

July 2025 – Present

- Machine Learning Model Development:** Designed, trained, and optimized supervised and unsupervised ML models (Regression, Classification, Clustering, Recommendation Systems) to solve business-critical problems.
- Deep Learning Applications:** Built an AI-powered advertising recommendation engine that uses LLMs to parse natural language queries, validates locations through semantic mapping, and ranks advertising spots using a weighted algorithm (footfall 30%, traffic 30%, POI density 20%, proximity 20%) with geospatial search across 290+ UAE categories, delivering optimized results within 5 seconds.
- Optimization & Hyperparameter Tuning:** Leveraged Grid Search, Random Search, and Bayesian Optimization to enhance model generalization and performance.
- MLOps & Deployment:** Deployed machine learning models into production using Docker, FastAPI, Flask, and cloud platforms (AWS, GCP, Azure) with CI/CD pipelines.

**IT Genics – Lahore**  
Associate AI Engineer

Jan 2024 – June 2025

- Developing AI Models:** Design, train, and deploy machine learning models to solve real-world challenges, including predictive analytics, natural language processing, and computer vision.
- Chatbot Development:** Build and implement intelligent chatbots for customer support, automation, and enhanced user experiences.
- No-Code Chatbot Platform:** Built a platform enabling companies to create custom chatbots without writing any code. Users input key details, and the system auto-generates fully functional chatbots, reducing deployment time and technical barriers.
- AI-VoiceBot:** Developed intelligent voicebot using twilio and Llama with NLP to perform desired functions.

**IT Solutions World Wide – Islamabad**  
Junior AI Engineer

Jul 2023 – Dec 2023

- Developed and deployed advanced LLMs, including GPT-3, Llama, BERT, and Falcon, using transfer learning and fine-tuning to address complex NLP challenges, enhancing the company's AI capabilities and enabling the development of customized chatbots for various use cases.
- Collaborated with cross-functional teams to gather requirements and design end-to-end AI pipelines, implementing algorithms by leveraging neural networks, deep learning frameworks (TensorFlow, PyTorch), and reinforcement learning to drive efficiency and innovation.

## EDUCATION

**NUCES, FAST – Islamabad**  
Bachelor of Computer Science

Aug 2019 – Jun 2023

## PROJECTS

### AITRAX – Pulmonary Fibrosis Progression Using Deep Learning

- Using TensorFlow and Keras, the model predicts and tracks disease progression with 72.56% accuracy by leveraging medical imaging data like CT scans for training deep learning models. The project's results have contributed to developing more effective treatment strategies for pulmonary fibrosis and enhancing patient outcomes.

### Cardiovascular Risk Prognostication and Classification System

- The model achieved a 96.82% accuracy rate in predicting heart disease likelihood, as assessed through cross-validation and comparison with actual medical outcomes. This success underscores the potential of machine learning in healthcare, suggesting that similar models could be developed for other diseases to enhance global health outcomes.

## CERTIFICATIONS

**IBM:** Tools for Data Science

**IBM:** Data Science Methodology

**IBM:** Databases and SQL for Data Science (Python)

**Microsoft:** Microsoft Azure Fundamentals (AZ-900)

**IBM:** Machine Learning with Python

**IBM:** Data Analysis With Python

## ADDITIONAL SKILLS

**Skills:** Python, Deep Learning, Neural Networks, Machine Learning, TensorFlow, Keras, PyTorch, Scikit-learn, Data Visualization, Data Analysis, SQL, NLP, Finetuning, CUDA, Text Generation, Transformer Models, Language Modeling, Azure, Database Management, Data Modeling, Fast APIs, VectorDBs.